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United States Patent
Kind Code
Date of Patent
Inventor(s)

12407092
B2
September 02, 2025
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Electronic device having antennas in ventilation ducts

Abstract

An electronic device may have an upper housing and a lower housing separated by a slot. A substrate may be mounted in the lower housing and may have an inner surface facing an interior of the lower housing and an outer surface facing the slot. Ventilation port openings may extend from the outer surface to the inner surface. Ribs in the substrate may separate the ventilation port openings. The ribs may include notches at the inner surface. An antenna may include conductive traces within multiple ventilation port openings and extending through the notches. The conductive traces may be offset from the outer surface. The outer surface of the substrate may be free from conductive material from the antenna. This may serve to maximize antenna efficiency while protecting the antenna from external forces and debris or other contaminants that pass through the slot during use of the electronic device over time.

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Appl. No.: 17/573524

Filed: January 11, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230223681 A1	Jul. 13, 2023

Publication Classification

Int. Cl.: H01Q1/44 (20060101); H01Q13/10 (20060101); H01Q1/22 (20060101)

U.S. Cl.:

CPC H01Q1/44 (20130101); H01Q13/10 (20130101); H01Q1/2266 (20130101)

Field of Classification Search

CPC: H01Q (1/44); H01Q (13/10); H01Q (1/2266); H01Q (1/2291)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8228239	12/2011	Lagnado et al.	N/A	N/A
8264412	12/2011	Ayala	343/702	H01Q 9/42
9450289	12/2015	Guterman et al.	N/A	N/A
9653777	12/2016	Guterman et al.	N/A	N/A
9800965	12/2016	Miehl et al.	N/A	N/A
10268236	12/2018	Guterman et al.	N/A	N/A
10476131	12/2018	Chang	N/A	H01Q 13/106
10566678	12/2019	Wu	N/A	H01Q 7/00
10591960	12/2019	Tani et al.	N/A	N/A
10608321	12/2019	Jiang et al.	N/A	N/A
10727590	12/2019	Barrera et al.	N/A	N/A
10978806	12/2020	Garrido Lopez et al.	N/A	N/A
11132030	12/2020	Huang	N/A	E05D 3/02

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2568530	12/2019	EP	N/A

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Background/Summary

BACKGROUND

(1) This relates generally to electronic devices and, more particularly, to wireless electronic devices with antennas.

(2) Electronic devices often include antennas. For example, cellular telephones, computers, and other devices often contain antennas for supporting wireless communications.

(3) It can be challenging to form electronic device antenna structures with desired attributes. In some wireless devices, the presence of conductive housing structures can influence antenna performance. Device size can also affect performance. It can be difficult to achieve desired performance levels in a compact device, particularly when the compact device has conductive housing structures. In addition, antenna structures can be susceptible to damage or contaminants

that detrimentally impact performance.

(4) It would therefore be desirable to be able to provide improved wireless circuitry for electronic devices.

SUMMARY

(5) An electronic device may have a metal housing. The metal housing may have an upper housing in which a component such as a display is mounted and a lower housing in which a component such as a keyboard is mounted. Hinges may be used to mount the upper housing to the lower housing for rotation about a rotational axis.

(6) A substrate may be mounted in the lower housing between upper and lower conductive housing walls. The substrate may have an inner surface that faces an interior of the lower housing. The substrate may have an outer surface that faces a slot between the upper and lower housings. The substrate may include ventilation port openings that extend from the outer surface to the inner surface. The substrate may include ribs that separate the ventilation port openings. Notches may be formed in the ribs at the inner surface.

(7) An antenna may be disposed on the substrate for radiating through the slot. The antenna may include conductive traces within multiple ventilation port openings. The conductive traces may pass through the notches in the ribs at the inner surface of the substrate. The conductive traces may be offset from the outer surface of the substrate. The outer surface of the substrate may be free from conductive material in the antenna. This may serve to maximize antenna efficiency while protecting the antenna from external forces and debris or other contaminants that pass through the slot during use of the electronic device over time.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of an illustrative electronic device such as a laptop computer in accordance with some embodiments.

(2) FIG. 2 is a schematic diagram of an illustrative electronic device with wireless circuitry in accordance with some embodiments.

(3) FIG. 3 is a diagram of an illustrative antenna in accordance with some embodiments.

(4) FIG. 4 is a diagram showing hinge and flexible printed circuit structures bridging a gap between upper and lower housings in a laptop computer of the type shown in FIG. 1 in accordance with some embodiments.

(5) FIG. 5 is a rear (exterior) view of a lower laptop computer housing having a ventilation system in accordance with some embodiments.

(6) FIG. 6 is a rear (exterior) perspective view of an illustrative ventilation port substrate that may serve as an antenna carrier for an electronic device in accordance with some embodiments.

(7) FIG. 7 is a cross-sectional side view of an illustrative ventilation port substrate having an antenna that extends along the wall of a notched rib in accordance with some embodiments.

(8) FIG. 8 is a front (interior) perspective view of an illustrative ventilation port substrate having an antenna that extends along the walls of multiple notched ribs for distributing the antenna across multiple ventilation ports in accordance with some embodiments.

(9) FIG. 9 is a cross-sectional side view of an illustrative ventilation port substrate having an antenna that extends along a horizontal rib and through a notched vertical rib in accordance with some embodiments.

(10) FIG. 10 is a front (interior) perspective view of an illustrative ventilation port substrate having an antenna on a horizontal rib and extending through multiple notched vertical ribs for distributing the antenna across multiple ventilation ports in accordance with some embodiments.

(11) FIGS. 11 and 12 are cross-sectional side views of an illustrative laptop computer (in closed-lid

and open-lid configurations) showing how the ventilation port substrate of FIGS. 5-10 may minimize exposure of an antenna on the substrate to external forces and contaminants in accordance with some embodiments.

DETAILED DESCRIPTION

(12) An electronic device such as electronic device **10** of FIG. **1** may contain wireless circuitry. For example, electronic device **10** may contain wireless communications circuitry that operates in long-range communications bands such as cellular telephone bands and wireless circuitry that operates in short-range communications bands such as the 2.4 GHz Bluetooth® or other wireless personal area network (WPAN) bands and the 2.4 GHz and 5 GHz Wi-Fi® band or other wireless local area network (WLAN) bands (sometimes referred to as IEEE 802.11 bands or wireless local area network communications bands). Device **10** may also contain wireless communications circuitry for performing near-field communications, communications at millimeter/centimeter wave frequencies, light-based wireless communications, satellite navigation system communications, or other wireless communications.

(13) Device **10** may be a handheld electronic device such as a cellular telephone, media player, gaming device, or other device, may be a laptop computer, tablet computer, or other portable computer, may be a desktop computer, may be a computer display, may be a display containing an embedded computer, may be a television or set top box, wireless base station, wireless access point, home entertainment console, portable speaker, gaming accessory, wristwatch device, head-mounted display device, or other wearable device, or may be other electronic equipment. Configurations in which device **10** has a rotatable lid as in a portable (e.g., laptop) computer are sometimes described herein as an example. This is, however, merely illustrative. Device **10** may be any suitable electronic equipment.

(14) As shown in the example of FIG. **1**, device **10** may have a housing such as housing **12**. Housing **12** may be formed from plastic, metal (e.g., aluminum), fiber composites such as carbon fiber, glass, ceramic, other materials, and combinations of these materials. Housing **12** or parts of housing **12** may be formed using a unibody construction in which housing structures are formed from an integrated piece of material. Multipart housing constructions may also be used in which housing **12** or parts of housing **12** are formed from frame structures, housing walls, and other components that are attached to each other using fasteners, adhesive, and other attachment mechanisms.

(15) Some of the structures in housing **12** may be conductive. For example, metal parts of housing **12** such as metal housing walls may be conductive. Other parts of housing **12** may be formed from dielectric material such as plastic, glass, ceramic, non-conducting composites, etc. To ensure that antenna structures in device **10** function properly, care should be taken when placing the antenna structures relative to the conductive portions of housing **12**.

(16) If desired, portions of housing **12** may form part of the antenna structures for device **10**. For example, conductive housing sidewalls may form all or part of an antenna ground. The antenna ground may include planar portions and/or portions that form one or more cavities for cavity-backed antennas. In addition to portions of housing **12**, the cavities in the cavity-backed antennas may be formed from metal brackets, sheet metal members, and other internal metal structures, and/or metal traces on dielectric structures (e.g., plastic structures) in device **10**. Metal traces may be formed on dielectric structures using molded interconnect device techniques (e.g., techniques for selectively plating metal traces onto regions of a plastic part that contains multiple shots of plastic with different affinities for metal), using laser direct structuring techniques (e.g., techniques in which laser light exposure is used to activate selective portions of a plastic structure for subsequent electroplating metal deposition operations), or using other metal trace deposition and patterning techniques.

(17) As shown in FIG. **1**, device **10** may have input-output devices such as track pad **18** (e.g., a touch pad, mouse, other touch-based user input device) and keyboard **16** (e.g., having a set of

mechanical and/or electronic-based keys and/or a touch screen display). Device **10** may also have components such as cameras, microphones, speakers, buttons, status indicator lights, buzzers, sensors, and other input-output devices. These devices may be used to gather input for device **10** and may be used to supply a user of device **10** with output. Connector ports in device **10** may receive mating connectors (e.g., an audio plug, a connector associated with a data cable such as a Universal Serial Bus cable, a data cable that handles video and audio data such as a cable that connects device **10** to a computer display, television, or other monitor, etc.).

(18) Device **10** may include a display such a display **14**. Display **14** may be a liquid crystal display (LCD), a plasma display, an organic light-emitting diode (OLED) display, an electrophoretic display, or a display implemented using other display technologies. A touch sensor may be incorporated into display **14** (e.g., display **14** may be a touch screen display) or display **14** may be insensitive to touch. Touch sensors for display **14** may be resistive touch sensors, capacitive touch sensors, acoustic touch sensors, light-based touch sensors, force sensors, or touch sensors implemented using other touch technologies.

(19) Device **10** may have a one-piece housing or a multi-piece housing. As shown in FIG. **1**, for example, electronic device **10** may be a device such as a portable computer or other device that has a two-part housing formed from an upper housing portion such as upper housing **12A** and a lower housing portion such as lower housing **12B**. Upper housing **12A** may include display **14** and may sometimes be referred to as a display housing or lid. Lower housing **12B** may sometimes be referred to as a base housing or main housing.

(20) Housings **12A** and **12B** may be connected to each other using hinge structures located along the upper edge of lower housing **12B** and the lower edge of upper housing **12A**. For example, housings **12A** and **12B** may be coupled by hinges **26** such as hinges **26A** and **26B** that are located at opposing left and right sides of housing **12** along a rotational axis such as axis **22** (sometimes referred to herein as hinge axis **22**). A slot-shaped opening such as opening **20** may be formed between upper housing **12A** and lower housing **12B** and may be bordered on either end by hinges **26A** and **26B**. Opening **20** may sometimes be referred to herein as gap **20** or slot **20** between upper housing **12A** and lower housing **12B**. Hinges **26A** and **26B**, which may be formed from conductive structures such as metal structures, may allow upper housing **12A** to rotate about axis **22** in directions **24** relative to lower housing **12B**. Slot **20** extends along the rear edge of lower housing **12B** parallel to axis **22**. The lateral plane of upper housing (lid) **12A** and the lateral plane of lower housing **12B** may be separated by an angle that varies between 0° when the lid is closed to 90°, 140°, 160°, or more when the lid is fully opened.

(21) A schematic diagram showing illustrative components that may be used in device **10** is shown in FIG. **2**. As shown in FIG. **2**, device **10** may include control circuitry such as control circuitry **30**. Control circuitry **30** may include storage and/or processing circuitry. Storage in control circuitry **30** may include hard disk drive storage, nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid-state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry **30** may be used to control the operation of device **10**. This processing circuitry may include one or more microprocessors, microcontrollers, digital signal processors, host processors, baseband processor integrated circuits, application specific integrated circuits, central processing units (CPUs), graphics processing units (GPUs), etc. Control circuitry **30** may be configured to perform operations in device **10** using hardware (e.g., dedicated hardware or circuitry), firmware, and/or software. Software code for performing operations in device **10** may be stored on control circuitry **30** (e.g., storage in control circuitry **30** may include non-transitory (tangible) computer readable storage media that stores the software code). The software code may sometimes be referred to as program instructions, software, data, instructions, or code. Software code stored on the storage may be executed by processing circuitry in control circuitry **30**.

(22) Control circuitry **30** may be used to run software on device **10** such as internet browsing

applications, voice-over-internet-protocol (VOIP) telephone call applications, email applications, media playback applications, operating system functions, etc. To support interactions with external equipment, control circuitry **30** may be used in implementing communications protocols. Communications protocols that may be implemented using control circuitry **30** include internet protocols, wireless local area network protocols (e.g., IEEE 802.11 protocols-sometimes referred to as WiFi®), protocols for other short-range wireless communications links such as the Bluetooth® protocol or other WPAN protocols, IEEE 802.11ad protocols, cellular telephone protocols, MIMO protocols, antenna diversity protocols, satellite navigation system protocols, antenna-based spatial ranging protocols (e.g., radio detection and ranging (RADAR) protocols or other desired range detection protocols for signals conveyed at millimeter and centimeter wave frequencies), etc. Each communication protocol may be associated with a corresponding radio access technology (RAT) that specifies the physical connection methodology used in implementing the protocol.

(23) Device **10** may include input-output devices **32**. Input-output devices **32** may be used to allow data to be supplied to device **10** and to allow data to be provided from device **10** to external devices. Input-output devices **32** may include user interface devices, data port devices, and other input-output components. For example, input-output devices may include touch screens, displays without touch sensor capabilities, buttons, joysticks, scrolling wheels, touch pads, key pads, keyboards, microphones, cameras, speakers, status indicators, light sources, audio jacks and other audio port components, digital data port devices, light sensors, accelerometers, proximity sensors, and other sensors and input-output components.

(24) Device **10** may include wireless communications circuitry **34** that allows control circuitry **30** of device **10** to communicate wirelessly with external equipment. The external equipment with which device **10** communicates wirelessly may be a computer, a cellular telephone, a watch, a router, access point, or other wireless local area network equipment, a wireless base station in a cellular telephone network, a display, a head-mounted device, or other electronic equipment. Wireless communications circuitry **34** may include radio-frequency (RF) transceiver circuitry **48** and one or more antennas such as antenna **40**. Configurations in which device **10** contains a single antenna may sometimes be described herein as an example. In general, device **10** may include any number of antennas.

(25) Transceiver circuitry **48** may support communications in Extremely High Frequency (EHF) or millimeter wave communications bands between about 30 GHz and 300 GHz, in centimeter wave communications bands between about 10 GHz and 30 GHz (sometimes referred to as Super High Frequency (SHF) bands), wireless local area network (WLAN) communications bands such as the 2.4 GHz and 5 GHz Wi-Fi® (IEEE 802.11) bands, wireless personal area network (WPAN) communications bands such as the 2.4 GHz Bluetooth® communications band, cellular telephone communications bands such as a cellular low band (LB) (e.g., 600 to 960 MHz), a cellular low-midband (LMB) (e.g., 1400 to 1550 MHz), a cellular midband (MB) (e.g., from 1700 to 2200 MHz), a cellular high band (HB) (e.g., from 2300 to 2700 MHz), a cellular ultra-high band (UHB) (e.g., from 3300 to 5000 MHz, or other cellular communications bands between about 600 MHz and about 5000 MHz (e.g., 3G bands, 4G LTE bands, 5G New Radio Frequency Range 1 (FR1) bands below 10 GHz, etc.), a near-field communications (NFC) band (e.g., at 13.56 MHz), satellite navigations bands (e.g., an L1 global positioning system (GPS) band at 1575 MHz, an L5 GPS band at 1176 MHz, a Global Navigation Satellite System (GLONASS) band, a BeiDou Navigation Satellite System (BDS) band, etc.), ultra-wideband (UWB) communications band(s) supported by the IEEE 802.15.4 protocol and/or other UWB communications protocols (e.g., a first UWB communications band at 6.5 GHz and/or a second UWB communications band at 8.0 GHz), and/or any other desired communications bands. The communications bands handled by the radio-frequency transceiver circuitry may sometimes be referred to herein as frequency bands or simply as “bands,” and may span corresponding ranges of frequencies. Transceiver circuitry **48** may include one or more integrated circuits, power amplifier circuitry, low-noise input amplifiers,

passive radio-frequency components, switching circuitry, transmission line structures, and other circuitry for handling radio-frequency signals.

(26) If desired, device **10** may be supplied with a battery such as battery **36**. Control circuitry **30**, input-output devices **32**, wireless communications circuitry **34**, and power management circuitry associated with battery **36** may produce heat during operation. To ensure that these components are cooled satisfactorily, device **10** may be provided with a cooling system such as cooling system **38**. Cooling system **38**, which may sometimes be referred to as a ventilation system, may include one or more fans and other equipment for removing heat from the components of device **10**. Cooling system **38** may include structures that form airflow ports (e.g., openings in ventilation port structures located along slot **20** of FIG. **1** or other portions of device **10** through which cool air may be drawn by one or more cooling fans and through which air that has been warmed from heat produced by internal components may be expelled). Airflow ports, which may sometimes be referred to as cooling ports, ventilation ports, air exhaust and entrance ports, etc., may be formed from arrays of openings in plastic ventilation port structures or other structures associated with cooling system **38**.

(27) Radio-frequency transceiver circuitry **48** and antenna(s) **40** may be used to handle one or more radio-frequency communications bands. For example, circuitry **48** may include wireless local area network transceiver circuitry that may handle a 2.4 GHz band for WiFi® and/or Bluetooth® communications and, if desired, may include 5 GHz transceiver circuitry (e.g., for WiFi®). If desired, transceiver circuitry **48** and antenna(s) **40** may handle communications in other bands (e.g., cellular telephone bands, near field communications bands, bands at millimeter wave frequencies, etc.).

(28) Transceiver circuitry **48** may convey radio-frequency signals using one or more antennas **40** (e.g., antennas **40** may convey the radio-frequency signals for the transceiver circuitry). The term “convey radio-frequency signals” as used herein means the transmission and/or reception of the radio-frequency signals (e.g., for performing unidirectional and/or bidirectional wireless communications with external wireless communications equipment). Antennas **40** may transmit the radio-frequency signals by radiating the radio-frequency signals into free space (or to free space through intervening device structures such as a dielectric cover layer). Antennas **40** may additionally or alternatively receive the radio-frequency signals from free space (e.g., through intervening devices structures such as a dielectric cover layer). The transmission and reception of radio-frequency signals by antennas **40** each involve the excitation or resonance of antenna currents on an antenna resonating element in the antenna by the radio-frequency signals within the frequency band(s) of operation of the antenna.

(29) Antennas **40** in wireless circuitry **34** may be formed using any suitable antenna structures. For example, antennas **40** may include antennas with resonating elements that are formed from stacked patch antenna structures, loop antenna structures, patch antenna structures, inverted-F antenna structures, slot antenna structures, planar inverted-F antenna structures, monopole antenna structures, dipole antenna structures, helical antenna structures, Yagi (Yagi-Uda) antenna structures, dielectric resonator antennas, hybrids of these designs, etc. If desired, one or more of antennas **40** may be cavity-backed antennas. Different types of antennas may be used for different bands and combinations of bands. If desired, antennas **40** may be arranged in one or more phased antenna arrays.

(30) As shown in FIG. **2**, transceiver circuitry **48** in wireless communications circuitry **34** may be coupled to antennas such as antenna **40** using radio-frequency transmission line paths such as transmission line **50**. Transmission line paths in device **10** such as transmission line **50** may include coaxial cables, microstrip transmission lines, stripline transmission lines, edge-coupled microstrip transmission lines, edge-coupled stripline transmission lines, waveguide transmission lines (e.g., coplanar waveguides, grounded coplanar waveguides, etc.), transmission lines formed from combinations of transmission lines of these types, etc.

(31) Transmission line paths in device **10** such as transmission line **50** may be integrated into rigid and/or flexible printed circuit boards if desired. In one suitable arrangement, transmission line paths in device **10** may include transmission line conductors (e.g., signal and/or ground conductors) that are integrated within multilayer laminated structures (e.g., layers of a conductive material such as copper and a dielectric material such as a resin that are laminated together without intervening adhesive) that may be folded or bent in multiple dimensions (e.g., two or three dimensions) and that maintain a bent or folded shape after bending (e.g., the multilayer laminated structures may be folded into a particular three-dimensional shape to route around other device components and may be rigid enough to hold its shape after folding without being held in place by stiffeners or other structures). All of the multiple layers of the laminated structures may be batch laminated together (e.g., in a single pressing process) without adhesive (e.g., as opposed to performing multiple pressing processes to laminate multiple layers together with adhesive). Filter circuitry, switching circuitry, impedance matching circuitry, and other circuitry may be interposed within the transmission lines, if desired.

(32) Transmission line **50** in device **10** may be coupled to antenna feed **42** of antenna **40**. Antenna **40** of FIG. 2 may, for example, form an inverted-F antenna, a planar inverted-F antenna, a slot antenna, a hybrid inverted-F slot antenna or other antenna having an antenna feed such as antenna feed **42** with a positive antenna feed terminal such as positive antenna feed terminal **44** and a ground antenna feed terminal such as ground antenna feed terminal **46**. Transmission line **50** may include a positive transmission line conductor **52** (sometimes referred to herein as signal conductor **52**) and a ground transmission line conductor **54** (sometimes referred to herein as ground conductor **54**). Signal conductor **52** may be coupled to positive antenna feed terminal **44** and ground conductor **54** may be coupled to ground antenna feed terminal **46**. Other types of antenna feed arrangements may be used (e.g., indirect feed arrangements, feed arrangements in which antenna **40** is fed using multiple feeds, etc.) and multiple antennas **40** may be provided in device **10**, if desired. The feeding configuration of FIG. 2 is merely illustrative.

(33) Filter circuitry, switching circuitry, impedance matching circuitry, and other circuitry may be interposed within transmission line **50**, in or between parts of antenna **40**, or in other portions of wireless communications circuitry **34**, if desired. Control circuitry **30** may be coupled to transceiver circuitry **48** and input-output devices **32**. During operation, input-output devices **32** may supply output from device **10** and may receive input from sources that are external to device **10**. Control circuitry **30** may use wireless communications circuitry **34** to transmit and receive wireless signals.

(34) FIG. 3 is a schematic diagram of an illustrative antenna for device **10**. In the example of FIG. 3, antenna **40** is an inverted-F antenna having inverted-F antenna resonating element **58** and antenna ground **56** (sometimes referred to herein as ground plane **56**, ground structures **56**, antenna ground structures **56**, or ground **56**). Antenna resonating element **58** (sometimes referred to herein as antenna radiating element **58**, resonating element **58**, or radiating element **58**) may have a main resonating element arm such as arm **60**. If desired, antenna resonating element **58** may have multiple branches (e.g., a first branch formed from arm **60**, a second branch formed from arm **60'**, etc.). The lengths of each of the branches of antenna resonating element **58** may be selected to support communications band resonances at desired frequencies (e.g., a high band resonance may be supported using a shorter branch such as arm **60'** and a low band resonance may be supported using a longer branch such as arm **60**). Antenna resonances may also be produced from resonating element harmonics and/or using parasitic antenna resonating elements.

(35) As shown in FIG. 3, antenna resonating element **58** (e.g., arm **60**) may be coupled to antenna ground **56** by return path **62**. Antenna feed **42** may be coupled between arm **60** and antenna ground **56** in parallel with return path **62**. Positive antenna feed terminal **44** may be coupled to arm **60**. Ground antenna feed terminal **46** may be coupled to antenna ground **56**. Antenna ground **56** may be formed from metal portions of housing **12** (e.g., portions of lower housing **12B** of FIG. 1), metal

traces on a printed circuit or other carrier/substrate, internal metal bracket members, sheet metal members, metal foil, and/or other conductive structures in device **10**. This example is merely illustrative and in general, antenna **40** may include an antenna resonating element having any desired shape and architecture.

(36) Metal traces on one or more flexible printed circuits may bisect slot **20** of FIG. **1**. Consider, for example, the illustrative configuration of device **10** that is shown in FIG. **4**. In the example of FIG. **4**, upper housing **12A** is separated from lower housing **12B** by air-filled slot **20**. Hinges **26A** and **26B** may be coupled between housings **12A** and **12B** along the respective left and right edges of device **10**. One or more flexible printed circuits such as flexible printed circuit **64** may bisect slot **20** along the length of slot **20**, thereby creating two slots (i.e., two separate slot-shaped portions of slot **20**) such as slots **20-1** and **20-2**. Flexible printed circuit **64** may contain one or more sheets of flexible dielectric substrate material such as a layer of polyimide or a sheet of other flexible polymers.

(37) Flexible printed circuit **64** may include signal lines **70** for routing display signals (i.e., data signals associated with displaying images on display **14** of FIG. **1**) and other signals (e.g., camera signals, backlight signals, power signals, touch sensor signals, etc.) between upper housing **12A** and lower housing **12B**. Ground traces **66** may be provided on the outer edges of flexible printed circuit **64** (i.e., in flexible printed circuit **64**, signal lines **70** may be flanked on opposing sides by ground traces **66**). Ground traces **66** may be formed from copper or other metal and may have any suitable widths (e.g., 1 mm to 3 mm, less than 1 mm, more than 1 mm, etc.). Ground traces **66** may be shorted to metal in housings **12A** and **12B** using screws, other fasteners, welds, conductive adhesive, solder, or other conductive coupling mechanism (see, e.g., conductive ground connections **68**).

(38) With this type of arrangement, slots (openings) **20-1** and **20-2** may be surrounded by metal. For example, slots **20-1** and **20-2** may be surrounded by metal portions of upper housing **12A** and lower housing **12B** on their top and bottom edges. Hinges **26A** and **26B** and ground traces **66** may also be formed from metal and may help define the shapes of slots **20-1** and **20-2**. As shown in FIG. **4**, slot **20-1** may have a left edge formed by hinge **26A** and an opposing right edge formed from the ground traces on flexible printed circuit **64**. Slot **20-2** may have a left edge formed from flexible printed circuit **64** and an opposing right edge formed from hinge **26-B**. The example of FIG. **4** in which one flexible printed circuit divides slot **20** into two separate slots is merely illustrative. If desired, two or more flexible printed circuits may divide slot **20** into three or more separate slots. Two or more separate flexible printed circuits may divide slot **20** into two separate slots **20-1** and **20-2** if desired (e.g., two or more separate flexible printed circuits may be interposed between slots **20-1** and **20-2**).

(39) During wireless operation of device **10**, slots **20-1** and **20-2** may serve as antenna apertures for respective electrically isolated antennas **40** in lower housing **12B** of device **10**. For example, a first antenna **40** may be mounted within lower housing **12B** and aligned with slot **20-1** and a second antenna **40** may be mounted within lower housing **12B** and aligned with slot **20-2**. Conductive structures in lower housing **12B** may form cavity structures for each of the antennas **40** (e.g., cavity-shaped ground structures or other ground structures that form part of antenna ground **56** of FIG. **3**). By aligning antennas **40** with separate slots between lower housing **12B** and upper housing **12A** in device **10**, the antennas may exhibit sufficient electrical isolation from each other (e.g., such that the antennas may be used to form a multiple-input-multiple-output (MIMO) antenna array at 2.4 GHz and/or 5 GHz and/or other suitable frequencies for wireless local area network communications, etc.).

(40) Device **10** may have ventilation port structures such as ventilation port structures **72** mounted along the rear edge of lower housing **12B** or elsewhere in device **10**. Ventilation port structures **72** may have arrays of openings that form ventilation ports. Fans in cooling system **38** (FIG. **2**) may be used to draw air into lower housing **12B** through the openings and may be used to exhaust air that

has been warmed by the circuitry in lower housing 12B through the openings. Separate ventilation port structures 72 may be aligned with slots 20-1 and 20-2 if desired. For example, a first ventilation port structure 72 may be interposed between the antenna 40 aligned with slot 20-1 and hinge 26A whereas a second ventilation port structure 72 is interposed between the antenna 40 aligned with slot 20-2 and hinge 26B. In another suitable arrangement, ventilation port structures 72 may be interposed between antennas 40 and flexible printed circuit 64. If desired, multiple antennas 40 may be aligned with slot 20-1 and/or multiple antennas 40 may be aligned with slot 20-2.

(41) An illustrative ventilation port structure of the type that may be formed along the rear edge of lower housing 12B is shown in the rear view of lower housing 12B of FIG. 5. In the example of FIG. 5, lower housing 12B has conductive upper wall 12B-1 and conductive lower wall 12B-2. The ventilation port structure in lower housing 12B (e.g., ventilation port structure 72 of FIG. 4) may include a ventilation port substrate 74. Ventilation port substrate 74 may be mounted along the rear edge of lower housing 12B between conductive upper wall 12B-1 and conductive lower wall 12B-2.

(42) Substrate 74 may have a top surface 84 facing conductive upper wall 12B-1 and may have an opposing bottom surface 86 facing conductive lower wall 12B-2 (e.g., substrate 74 may be disposed within the interior of lower housing 12B such that top surface 84 is at or adjacent to conductive upper wall 12B-1 and bottom surface 86 is at or adjacent to conductive lower wall 12B-2). Substrate 74 may have an exterior (rear) facing surface such as rear surface 82 that faces the exterior of lower housing 12B. Substrate 74 may also have an opposing interior (front facing) surface opposite rear surface 82. Rear surface 82 may sometimes be referred to herein as rear-facing surface 82, exterior-facing surface 82, exterior surface 82, rear face 82, or exterior face 82. Substrate 74 may be formed from plastic, ceramic, polyimide, or any other desired dielectric materials.

(43) Ventilation port substrate 74 may have ventilation port openings 76. Openings 76 may be used to allow air to enter and exit the interior of lower housing 12B. There may be one or more openings 76 associated with each airflow entrance location and each airflow exit location in substrate 74. Openings 76 may, if desired, be arranged in arrays (ventilation ports) 78 such as a first array 78-1 and a second array 78-2 (e.g., arrays of 6-20 openings, more than 4 openings, fewer than 30 openings, etc.). One or more flexible printed circuits (e.g., flexible printed circuit 64 of FIG. 4) may be interposed between two or more opening(s) 76 (e.g., at location 80 between arrays 78-1 and 78-2 as shown in FIG. 5).

(44) FIG. 6 is a rear perspective view of a portion of substrate 74 showing how openings 76 may be provided in substrate 74 to allow air to flow into and out of device 10. In the example of FIG. 6, openings 76 have rectangular outlines. This is merely illustrative. Any suitable shapes (circular, oval, triangular, etc.) may be used in forming openings 76.

(45) As shown in FIG. 6, substrate 74 may have a front (interior) facing surface such as front surface 88 that faces the interior of lower housing 12B (FIG. 5) when substrate 74 is mounted within lower housing 12B. Front surface 88 may sometimes be referred to herein as front-facing surface 88, interior-facing surface 88, interior surface 88, front face 88, or interior face 88. Openings 76 may extend from rear surface 82 to front surface 88 through substrate 74. Openings 76 are cavities that are generally free from material (e.g., plastic from substrate 74) to allow for the passage of air through openings 76. If desired, two or more openings 76 at rear surface 82 may merge to a single shared opening at front surface 88 of substrate 74.

(46) Substrate 74 may include ribs 90 that separate each opening 76 from one or more other openings 76. In the example of FIG. 6, substrate 74 includes two rows of openings 76 that are separated by a horizontal rib 90H. This is merely illustrative and, in general, openings 76 may be arranged in any desired pattern. Substrate 74 also includes vertical ribs 90V that separate each opening 76 from a laterally adjacent opening 76 in its row. Vertical ribs 90V may sometimes be

referred to herein as vertical walls **90V** (e.g., vertical walls defining the vertical left/right edges of openings **76**), sidewalls **90V** (e.g., sidewalls of openings **76**), or simply as walls **90V**. Similarly, horizontal ribs **90H** may sometimes be referred to herein as horizontal walls **90H** (e.g., horizontal walls defining the horizontal top/bottom edges of openings **76**) or simply as walls **90H**.

(47) Antenna structures may be supported by one or more dielectric support structures in device **10**. For example, metal traces, metal foil, sheet metal structures, or other conductive structures for antenna(s) **40** may be supported on plastic structures or other dielectric structures. With one illustrative arrangement which may sometimes be described herein as an example, some or all of the conductive structures that form antenna(s) **40** (e.g., portions of antenna resonating element **58** and/or antenna ground **56** of FIG. **3**, feed and return path structures, etc.) may be formed on a plastic structure such as substrate **74** of FIGS. **5** and **6** that is also used for forming some or all of a ventilation system (e.g., cooling system **38** of FIG. **2**) for device **10**.

(48) Since rear surface **82** of substrate **74** faces the exterior of device **10** while mounted within lower housing **12B** (FIG. **5**), disposing antenna **40** (e.g., antenna resonating element **58** of FIG. **3**) on rear surface **82** (e.g., at location **92** of FIG. **6**) may allow antenna **40** to freely convey radio-frequency signals with external equipment with maximum antenna efficiency. When mounted at location **92**, some of the antenna may extend into one or more of openings **76** such as at location **94**. This may serve to increase the resonant length of the antenna (e.g., for maximizing antenna efficiency at lower frequencies).

(49) However, disposing antenna **40** on rear surface **82** of substrate **74** may expose antenna **40** to external forces and contaminants that may damage antenna **40** and/or that may detune or deteriorate the performance of antenna **40** over time. For example, when device **10** is in a closed-lid configuration and placed rear-first vertically onto a surface (e.g., when device **10** is placed within a messenger bag, purse, satchel, or other portable container), the surface may impact device **10** in a way that causes excessive force to be applied at rear surface **82** of substrate **74** and thus to an antenna **40** at location **92**. These forces may damage the antenna and may deteriorate antenna performance over time. In addition, debris, dust, or items located on the surface may become lodged within lower housing **12B** at or adjacent to rear surface **82** and can undesirably block or damage an antenna **40** at location **92**.

(50) To help mitigate these external factors, an entirety the antenna resonating element for the antenna (or an entirety of antenna **40**) may be located within a corresponding ventilation duct **76**, such as at location **96** of FIG. **6**. In these configurations, no part of the antenna resonating element (or no part of antenna **40**) is located on rear surface **82** (e.g., rear surface **82** may be free from conductive material in the antenna resonating element or antenna **40**). However, substrate **74** may be relatively narrow (e.g., as measured from rear surface **82** to front surface **88**) to minimize device form factor and to maximize the amount of space in lower housing **12B** (FIG. **1**) that is available for other device components. As such, opening **76** may not be sufficiently long on its own to accommodate the entire length of the antenna resonating element in antenna **40**, particularly for covering relatively low frequencies.

(51) To optimize the radio-frequency performance of antenna **40** across all frequencies of interest while minimizing the effect of external forces and contaminants on antenna **40**, antenna **40** may be disposed within (e.g., distributed between) multiple openings **76**. Substrate **74** may include structures at front surface **88** that allow antenna **40** to extend between multiple openings **76** at the interior-facing side of substrate **74** without exposing the antenna at rear surface **82**. These structures may include notches or recesses in the ribs of substrate **74** that accommodate antenna **40** as the antenna extends between multiple openings **76**. Part of the antenna **40** may be patterned directly onto the wall(s) of the notches or recesses in the ribs of substrate **74** and/or the antenna **40** may pass through the notches or recesses without being patterned onto the walls of the notches or recesses.

(52) FIG. **7** is a cross-sectional side view (e.g., as taken along line AA' of FIG. **6**) showing how

substrate **74** may include a notch or recess at front surface **88** for accommodating antenna **40**. As shown in FIG. 7, substrate **74** may include a first opening **76-1** (e.g., from the first row of openings **76** in FIG. 6) and a second opening **76-2** (e.g., from the second row of openings **76** in FIG. 6) beneath first opening **76-1**.

(53) In the example of FIG. 7, opening **76-1** and opening **76-2** form two respective ventilation port openings at rear surface **82** of substrate **74** but merge together to form a single opening at front surface **88** of substrate **74**. As such, opening **76-1** extends along a first longitudinal axis **98** and opening **76-2** extends along a second longitudinal axis **100** that is oriented non-parallel with respect to first longitudinal axis **98**. Opening **76-1** may exhibit symmetry about longitudinal axis **98** and opening **76-2** may exhibit symmetry about longitudinal axis **100** if desired. Rear surface **82** may be oriented non-parallel with respect to front surface **88** if desired.

(54) Opening **76-1** is separated from opening **76-2** (at rear surface **82**) by horizontal rib **90H** of substrate **74**. In the example of FIG. 7, horizontal rib **90H** has a chevron-shaped profile (e.g., in the Y-Z plane). Horizontal rib **90H** has a first surface that forms a curved (e.g., concavely curved) portion of the rear surface **82** of substrate **74** and has an opposing second surface **104** that faces the front face of substrate **74**. Surface **104** of horizontal rib **90H** may form the bottom (lower) edge, surface, or wall of opening **76-1** and may form the top (upper) edge, surface, or wall of opening **76-2**. Opening **76-1** may have a top (upper) edge **102** opposite surface **104** of horizontal rib **90H**. Opening **76-2** may have a bottom (lower) edge **106** opposite surface **104** of horizontal rib **90H**. The example of FIG. 7 is merely illustrative and, in general, openings **76-1** and **76-2** may have other shapes or profiles as the openings extend from rear surface **82** to front surface **88**. The openings need not merge at front surface **88**.

(55) As shown in FIG. 7, substrate **74** may include a vertical rib **90V** that extends perpendicular to horizontal rib **90H**. For example, horizontal rib **90H** may have a longitudinal axis that extends parallel to the X-axis (e.g., to separate the first and second rows of openings **76** as shown in FIG. 6) whereas vertical rib **90V** extends within the Y-Z plane of FIG. 7. Vertical rib **90V** may separate openings **76-1** and **76-2** from third and fourth openings located farther in the -X direction of FIG. 7 than openings **76-1** and **76-2**.

(56) Vertical rib **90V** may have a notch (recess) **108** at front surface **88**. Notch **108** may extend into opening **76-1** by distance **120** from an open end to an opposing closed end at rear wall **110** of notch **108** (e.g., rear wall **110** may form a portion of front surface **88** that is offset from the inner-most portion of front surface **88** by notch **108**). Distance **120** may be 1-6 mm, as an example. Notch **108** may have an upper edge **109** and a lower edge **111** that extends from the open end to rear wall **110** (sometimes referred to herein as notch wall **110** or notch rear wall **110**). Upper edge **109** may extend parallel to or non-parallel to lower edge **111**. Upper edge **109** and lower edge **111** need not be linear and may, if desired, have a curved shape, stepped shape, or other shapes. The example of FIG. 7 in which notch **108** extends only into opening **76-1** is merely illustrative. If desired, notch **108** may extend into both openings **76-1** and **76-2** or only into opening **76-2**. Rear wall **110** of notch **108** may be located within the merged portion of openings **76-1** and **76-2** (e.g., as shown in FIG. 7), may be vertically interposed between surface **104** of horizontal rib **90H** and top edge **102** of opening **76-1**, or may be vertically interposed between surface **104** of horizontal rib **90H** and bottom edge **106** of opening **76-2**.

(57) As shown in FIG. 7, antenna **40** may be disposed within opening **76-1**. Antenna **40** may include conductive trace **114** on surface **104** of horizontal rib **90H**, conductive trace **114** on the vertical surface of vertical rib **90V**, and conductive trace **118** on rear wall **110** of notch **108**. Conductive traces **114**, **116**, and **118** may form all or part of antenna resonating element **58** (e.g., arm **60**) and/or antenna ground **56** of FIG. 3. Conductive trace **116** may couple conductive trace **114** to conductive trace **118**. Conductive trace **118** may couple the conductive traces of antenna **40** within opening **76-1** to conductive traces in antenna **40** that are located within the opening in substrate **74** located on the side of vertical rib **90V** opposite to opening **76-1** (e.g., conductive trace

118 may couple conductive trace **116** to additional conductive traces on the opposing side of vertical rib **90V** and/or on the portion of horizontal rib **90H** located within the opening in substrate **74** located on the side of vertical rib **90V** opposite to opening **76-1**).

(58) When disposed in this way, conductive trace **114** may be laterally separated (offset) from rear surface **82** of substrate **74** by distance **112**. Distance **112** may be adjusted to tune the performance of antenna **40**. For example, shorter distances **112** may place antenna **40** closer to the exterior of device **10**, thereby increasing antenna efficiency, whereas longer distances **112** may better protect antenna **40** from external forces and contaminants, thereby extending the overall longevity of the antenna. Distance **112** may be selected to optimize these two factors (e.g., distance **112** may be between around 0.1-10 mm). The example of FIG. 7 is merely illustrative. If desired, antenna **40** may be disposed entirely within opening **76-2** or may be distributed between openings **76-1** and **76-2** (e.g., conductive trace **114** may extend onto the portion of surface **104** that defines an edge of opening **76-2**).

(59) FIG. 8 is an interior perspective view (e.g., as viewed in the -Y direction of FIG. 7) showing how antenna **40** may be distributed across multiple openings **76** in substrate **74**. As shown in FIG. 8, front surface **88** of substrate **74** faces the interior of lower housing **12B** when substrate **74** is mounted within lower housing **12B**. Antenna **40** may extend across at least three laterally-adjacent openings **76** in substrate **74** (e.g., openings **76-1**, **76-3**, and **76-4** within a first row of openings in substrate **74**). This is merely illustrative and, if desired, antenna **40** may extend across only two openings **76**, across more than three openings **76**, across openings in more than one row of openings, etc.

(60) Substrate **74** may include a first vertical rib **90V-1** that separates opening **76-1** from opening **76-3** and may include a second vertical rib **90V-2** that separates opening **76-3** from opening **76-4**. Vertical rib **90V-1** and vertical rib **90V-2** may each include a respective notch **108** at front surface **88** of substrate **74** (e.g., where each notch **108** has edges **111** and **109** extending from the interior-most portion of front surface **88** to a respective rear wall **110** formed from a portion of the vertical rib). Horizontal rib **90H** may separate openings **76-1**, **76-3**, and **76-4** from the second row of openings in substrate **74** (e.g., at the rear surface of substrate **74**).

(61) Antenna **40** may include conductive traces **114** located (patterned) on surface **104** of horizontal rib **90H** within openings **76-1**, **76-3**, and **76-4**. Antenna **40** may also include conductive traces **116** located on the surface of vertical rib **90V-1** defining an edge of opening **76-1**, on the surface of vertical rib **90V-1** defining an edge of opening **76-3**, on the surface of vertical rib **90V-2** defining an edge of opening **76-3**, and on the surface of vertical rib **90V-4** defining an edge of opening **76-4**. Antenna **40** may further include conductive traces **118** located on rear walls **110** of the notches **108** in vertical ribs **90V-1** and **90V-2**. In other words, antenna **40** (e.g., the antenna resonating element or resonating element arm for antenna **40**) may extend from opening **76-1**, through notch **108** in vertical rib **90V-1**, through opening **76-2**, through notch **108** in vertical rib **90V-2**, and into opening **76-3**. If desired, conductive traces **116** (e.g., on vertical rib **90V-2**) may be coupled to terminals **122** on top surface **84** of substrate **74**. Terminals **122** may alternatively be located on bottom surface **86** of substrate **74**. Terminals **122** may be used to form one or both of positive antenna feed terminal **44** and ground antenna feed terminal **46** and/or to form a return path **62** (FIG. 3) for antenna **40**. The radio-frequency transmission line for antenna **40** may be coupled to one or both of terminals **122**. If desired, additional conductive traces may be patterned on top surface **84** and/or bottom surface **86** of substrate **74** to form part of the transmission line for antenna **40** and/or for forming part of the antenna ground for antenna **40**.

(62) In this way, antenna **40** may be distributed across multiple openings **76** in substrate **74** to maximize the length of the antenna resonating element and thus antenna efficiency at relatively low frequencies. Separating antenna **40** from the rear surface of substrate **74** in this way may prevent antenna **40** from being subject to destabilizing external forces and contaminants. Forming notches **108** in vertical ribs **90V** at the front surface **88** of substrate **74** may allow antenna **40** to pass

between openings **76** without exposing any part of the antenna at the rear surface of substrate **74**. Patterning part of antenna **40** onto rear walls **110** of notches **108** (e.g., at distance **120** from the inner-most portion of front surface **88** as shown in FIG. 7) may allow an entirety of the antenna resonating element to be located as close to the rear face of substrate **74** as possible (e.g., to maximize antenna efficiency) while also optimizing the radiation pattern for the antenna relative to scenarios where notches **108** are omitted.

(63) The example of FIG. 8 is merely illustrative. The conductive traces used to form antenna **40** may have any desired shape. If desired, the conductive traces may pass between openings **76-1**, **76-3**, and **76-4** and through notches **108** without being patterned onto rear wall **110** of the notches. FIG. 9 is a cross-sectional side view showing how antenna **40** may be disposed in opening **76-1** for passing through notch **108** without being patterned onto the rear wall of the notch.

(64) As shown in FIG. 9, notch **108** may extend by distance **124** into opening **76-1** (e.g., where distance **124** is greater than distance **120** of FIG. 7). Distance **124** may be 3-9 mm, for example. Distance **124** and distance **122** of FIG. 7 may sometimes be referred to herein as the length of notch **108**. This example is merely illustrative. If desired, notch **108** may extend into both openings **76-1** and **76-2** or into only opening **76-2**. Openings **76-1** and **76-2** need not be merged at front surface **88**. Distance **124** may be sufficiently far so as to allow conductive trace **114** of antenna **40** to extend along surface **104** of horizontal rib **90H** into the adjacent opening parallel to the X axis without extending up onto any portion of the vertical surface(s) of vertical rib **90V**. In other words, conductive trace **114** may pass through notch **108** without being patterned onto any of the surfaces of vertical rib **90V**.

(65) FIG. 10 is an interior perspective view showing how antenna **40** may include conductive traces **114** on horizontal rib **90H** that pass from opening **76-1** through opening **76-3** and into opening **76-4** through notches **108** in vertical ribs **90V-1** and **90V-2** without being patterned onto any part of vertical ribs **90V-1** and **90V-2**. If desired, conductive traces may be patterned on one or more vertical ribs to couple conductive traces **114** to terminals **122**. Disposing antenna **40** in this way may further increase its antenna efficiency (e.g., by pushing more of the antenna closer to the rear surface of substrate **74**) and/or may simplify the manufacturing complexity of device **10** relative to the arrangement of FIGS. 7 and 8. Use of conductive traces on notches **108** may simplify the manufacturing complexity of substrate **74**, may reduce the volume occupied by substrate **74**, may reduce the weight of device **10**, may maximize the mechanical reliability of antenna **40**, and may maximize the antenna efficiency of antenna **40** relative to scenarios where conductive vias are used to couple conductive traces in different ventilation ports together.

(66) The example of FIG. 10 is merely illustrative. Antenna **40** may extend into any desired number of openings **76** in any desired number of rows of openings. If desired, a combination of the arrangement of FIGS. 7 and 8 and the arrangement of FIGS. 9 and 10 may be used (e.g., some portions of antenna **40** may be patterned onto rear wall **110** of the notch **108** in one or more vertical ribs **90V** whereas other portions of antennas **40** pass through the notch **108** in one or more vertical ribs **90V** without being patterned onto the corresponding rear wall **110**). Put differently, the notches **108** in vertical ribs **90V** may have different lengths if desired.

(67) FIG. 11 is a cross-sectional side view of device **10** in the vicinity of the rear edge of lower housing **12B** when upper housing **12A** is in a closed position (sometimes referred to herein as a closed lid configuration) after rotating upper housing **12A** about rotational axis **130** (e.g., axis **22** of FIG. 1). As shown in FIG. 11, the lateral surface of conductive upper wall **12B-1** may extend parallel or substantially parallel (e.g., within 30 degrees) to the lateral surface of conductive lower wall **12B-2**. Conductive upper wall **12B-1** and conductive lower wall **12B-2** may define the interior of lower housing **12B**. A main logic board, battery **36** (FIG. 2), a set of input-output devices **32**, cooling system **38**, transceiver circuitry **48**, control circuitry **30**, and other desired components may be mounted within the interior of lower housing **12B**. Substrate **74** may be mounted within the interior of lower housing **12B** between conductive upper wall **12B-1** and conductive lower wall

12B-2. By mounting substrate **74** in this way, an entirety of the antenna resonating element and substrate **74** may be interposed between conductive upper wall **12B-1** and conductive lower wall **12B-2** within the interior of lower housing **12B**. This may, for example, hide antenna **40** from view of a user at the exterior of device **10** and may protect antenna **40** from contaminants or damage. (68) Components such as keyboard **16** and track pad **18** (FIG. **1**) may operate through openings in conductive upper wall **12B-1**. Conductive lower wall **12B-2**, which may be joined to conductive upper wall **12B-1** around the lateral periphery of lower housing **12B** (e.g., such that conductive material surrounds the interior cavity and thus substrate **74**), may have feet or other support structures that allow device **10** to rest on a tabletop, a user's lap, or other support structure during operation. Fans and other cooling system structures (cooling system **38** of FIG. **2**) may be mounted within the interior of lower housing **12B**. Conductive upper wall **12B-1** may be electrically coupled to conductive lower wall **12B-2** using conductive structure **134**. Conductive structure **134** may be a conductive gasket formed from conductive foam, conductive fabric, a solid elastomeric conductive material, or other conductive material.

(69) An opening may be provided in conductive structure **134** to form an air passageway (ventilation opening) that is aligned with the openings **76** in substrate **74**. When device **10** is being used in this way, air may flow in and out of ventilation port structure **72** (FIG. **4**) through openings **76** in substrate **74**, as shown by arrows **136**. Openings **76** in substrate **74** (e.g., openings **76-1** and **76-2** of FIG. **11**) may allow (intake) air to pass from the right of substrate **74** to the left of substrate **74** and/or may allow (exhaust) air to pass from the left of substrate **74** to the right of substrate **74**. In the closed lid configuration, conductive material in upper housing **12A** may be separated from conductive material in lower housing **12B** by slot **132**. Antenna **40** may convey radio-frequency signals through slot **132**. Disposing antenna **40** within openings **76** and displaced from the rear surface of substrate **74** may protect antenna **40** from force and contaminants received through slot **132** (e.g., when device **10** is placed rear-edge down in a bag or on a surface).

(70) FIG. **12** shows device **10** in an illustrative lid-open configuration in which upper housing **12A** has been rotated into an open position about rotational axis **130**. Conductive upper wall **12B-1** may be separated from conductive material in upper housing **12A** by slot **138** whereas conductive lower wall **12B-2** is separated from conductive material in upper housing **12A** by slot **132**. Antenna **40** may convey radio-frequency signals through slots **138** and **132**. The examples of FIGS. **11** and **12** in which antenna **40** is disposed within opening **76-2** is merely illustrative. If desired, antenna **40** may be disposed within opening **76-1** or within both openings **76-1** and **76-2**. Antenna **40** may extend across multiple openings as shown in the examples of FIGS. **7-10**.

(71) Device **10** may gather and/or use personally identifiable information. It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

(72) The foregoing is merely illustrative and various modifications can be made by those skilled in the art without departing from the scope and spirit of the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. An electronic device comprising: a first conductive housing wall; a second conductive housing wall; a substrate disposed between the first and second conductive housing walls, wherein the substrate includes a first ventilation port opening, a second ventilation port opening, a rib that separates the first ventilation port opening from the second ventilation port opening, and a notch having a closed end formed from a portion of the rib and an open end opposite the closed end; and

an antenna radiating element on the substrate and extending, through the notch, from the first ventilation port opening to the second ventilation port opening, wherein the antenna radiating element comprises a conductive trace on the portion of the rib that forms the closed end of the notch.

2. The electronic device of claim 1, wherein the substrate comprises a plastic substrate.

3. The electronic device of claim 1, further comprising: a lower housing portion that includes the first and second conductive housing walls; an upper housing portion coupled to the lower housing portion by a hinge; and a display in the upper housing portion, wherein the antenna radiating element is configured to radiate through a slot between the lower housing portion and the upper housing portion.

4. The electronic device of claim 3, wherein the substrate has a first surface facing the slot and a second surface opposite the first surface, the first and second ventilation port openings extend through the substrate from the first surface to the second surface, and the antenna radiating element is separated from the second surface by a non-zero distance within the notch.

5. The electronic device of claim 4, wherein none of the antenna radiating element is located on the first surface.

6. The electronic device of claim 4, wherein the notch extends from the second surface towards the first surface.

7. The electronic device of claim 1, wherein the substrate comprises an additional rib that extends orthogonal to the rib and that separates the first ventilation port opening from a third ventilation port opening in the substrate.

8. The electronic device of claim 7, wherein the additional rib separates the first ventilation port opening from the second ventilation port opening at a first face of the substrate, the substrate has a second face opposite the first face, and the first ventilation port opening is merged with the second ventilation port opening at the second face.

9. An electronic device comprising: a housing having an upper housing portion that contains a display and having a lower housing portion, wherein the lower housing portion has opposing first and second conductive walls; a hinge that couples the upper housing portion to the lower housing portion, wherein the upper housing portion is rotatable with respect to the lower housing portion and is separated from the lower housing portion by at least one slot; a substrate disposed in the lower housing portion between the first and second conductive walls, wherein the substrate has a first surface that faces an interior of the lower housing portion and a second surface that faces the at least one slot; first and second ventilation ports in the substrate and extending from the second surface towards the first surface; and an antenna configured to radiate through the at least one slot and having a first conductive trace in the first ventilation port, a second conductive trace in the second ventilation port, and a third conductive trace that couples the first conductive trace to the second conductive trace, wherein the first, second, and third conductive traces are offset from the second surface of the substrate.

10. The electronic device of claim 9, wherein the substrate comprises: a wall that separates the first ventilation port from the second ventilation port; and a recess in the wall at the first surface of the substrate.

11. The electronic device of claim 10, wherein the recess has an open end, a rear wall opposite the open end, and an edge that extends from the open end to the rear wall, the third conductive trace being disposed on the edge of the recess.

12. The electronic device of claim 10, the substrate further comprising: an additional wall that forms an edge of the first ventilation port and that forms an edge of the second ventilation port, wherein the first, second, and third conductive traces are disposed on the additional wall and the third conductive trace extends through the recess.

13. The electronic device of claim 12, wherein the additional wall is oriented perpendicular to the wall.

14. The electronic device of claim 10, wherein the antenna comprises: a positive antenna feed terminal; and a fourth conductive trace on the wall that couples the first conductive trace to the positive antenna feed terminal.

15. An electronic device comprising: a housing having an upper housing portion that contains a display and having a lower housing portion, wherein the lower housing portion has opposing first and second conductive walls and the lower housing portion is separated from the upper housing portion by at least one slot; a substrate disposed in the lower housing portion between the first and second conductive walls, wherein the substrate has a first surface that faces an interior of the lower housing portion and a second surface that faces the at least one slot; first and second ventilation ports in the substrate that extend from the second surface towards the first surface, wherein the first and second ventilation ports are separated by a rib of the substrate at the second surface; a notch in the substrate, wherein the notch has a closed end defined by a portion of the rib, a second end opposite the first end, and an edge that extends from the closed end to the open end; and an antenna radiating element that is configured to radiate through the at least one slot and that includes a first conductive trace in the first ventilation port, a second conductive trace in the second ventilation port, and a third conductive trace on the edge of the notch, wherein the third conductive trace couples the first conductive trace to the second conductive trace.

16. The electronic device of claim 15, wherein the closed end of the notch is separated from the first surface by a first non-zero distance and is separated from the second surface by a second non-zero distance greater than the first non-zero distance.

17. The electronic device of claim 15, wherein the first, second, and third conductive traces are offset from the second surface of the substrate.

18. The electronic device of claim 17, wherein the second surface of the substrate is free of conductive material from the antenna radiating element.

19. The electronic device of claim 15, wherein the first and second ventilation ports are merged at the first surface of the substrate by the open end of the notch.

20. The electronic device of claim 15, wherein the first conductive wall is parallel to the second conductive wall, the edge of the notch being oriented at a non-parallel and non-perpendicular angle with respect to the first and second conductive walls.
